

Quiz 3

Question 1: Translate the sentence into predicate logic

Every car with four wheels is a vehicle.	Predicates: $C(x)$ = x is a car, $F(x)$ = x has four wheels, $V(x)$ = x is a vehicle $\forall x [(C(x) \wedge F(x)) \rightarrow V(x)]$
All mammals that live in water are whales or dolphins.	Predicates: $M(x)$ = x is a mammal, $W(x)$ = x lives in water, $Wh(x)$ = x is a whale, $D(x)$ = x is a dolphin $\forall x [(M(x) \wedge W(x)) \rightarrow (Wh(x) \vee D(x))]$

Question 2: Translate the sentence using nested quantifiers

"For each student, there is a teacher t who mentors s."	Let $M(t, s)$ = t mentors s $\forall s \exists t (M(t, s))$
"There exists a book b that every reader r has read."	Let $Read(r, b)$ = r has read b $\exists b \forall r (Read(r, b))$
"For every city c, there is a road r connecting c to another city."	Let $Connects(r, c, d)$ = road r connects city c to city d $\forall c \exists r \exists d ((c \neq d) \wedge Connects(r, c, d))$

Question 3: Negate the statement

$\forall x (\neg V(x) \vee W(x))$	$\exists x (V(x) \wedge \neg W(x))$
$\exists x (P(x) \wedge Q(x) \wedge \neg R(x))$	$\forall x (\neg P(x) \vee \neg Q(x) \vee R(x))$

Question 4 : "Our New World" with the given people, cities, items, and foods, together with the facts provided:

People: Anna, Ben

Cities: London, Rome

Items: Book, Hat

Foods: Pizza, Sushi

Facts:

1. Anna visited London, picked up a Book, and ate Pizza.
2. Anna visited Rome, picked up a Hat, and ate Pizza.
3. Ben visited London, picked up a Book, and ate Sushi.
4. Ben did not visit Rome.

Let us define the predicates clearly (using the most natural symbols from your list):

- $V(p, c)$: person p visited city c
- $P(p, i)$: person p picked up item i
- $A(p, f)$: person p ate food f

Domain of people = {Anna, Ben}

Domain of cities = {London, Rome}

Domain of items = {Book, Hat}

Domain of foods = {Pizza, Sushi}

Convert it to simple English and provide the truth value of the statement.

Statements	English	Truth value

$\forall p \exists c \exists i [V(p, c) \wedge P(p, i)]$	Every person visited some city and picked up some item.	T
$\exists p \exists c \exists f [V(p, c) \wedge A(p, f) \wedge (f = \text{Sushi})]$	Some person visited some city and ate Sushi.	T
$\forall c \exists p [V(p, c) \rightarrow A(p, \text{Pizza})]$	For every city, If anyone visited it, they ate Pizza.	F
$\forall p \forall c [V(p, \text{Rome}) \rightarrow A(p, \text{Pizza})]$	Every person who visited Rome ate Pizza.	T
$\exists p \neg \exists f A(p, f)$	There exists a person who ate no food at all.	F
$\exists p \exists c \exists i \exists f [V(p, c) \wedge P(p, i) \wedge A(p, f)]$	Some person visited a city, picked up an item, and ate food.	T